CS 516 Project

Fairness or Bias of LLMs in High-Stakes Domains: Hiring & Recidivism

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Motivation: Why Fairness in LLMs Matters

High-Stakes AI Decisions

LLMs increasingly used for hiring, resume screening, candidate ranking, job ad generation, and recidivism analysis. These decisions impact livelihood, freedom, and social mobility.

Inherited Biases

LLMs inherit historical, racial, gender, cultural, disability, and socioeconomic biases from training data.

Real Harm

Unfair AI causes lost jobs, discrimination, unjust incarceration, and violation of fundamental rights.

Urgent Research

Need systematic audit of LLM fairness before deployment in high-stakes domains.

Research Questions & Objectives

RQ1: How LLMs Exhibit Bias

Identify and characterize bias in hiring and recidivism prediction tasks across different demographic groups.

RQ2: Detection & Measurement

Explore methods to detect, measure, and quantify biases in LLM outputs and decision-making.

RQ3: Fairness Metrics

Evaluate LLM fairness using standard fairness metrics: Statistical Parity, Equal Opportunity, Disparate Impact.

Systematic Literature Review Methodology

Research Foundation

Comprehensive literature review of 10 high-quality peer-reviewed papers from AIES, FAccT, and NeurIPS conferences on LLM bias and fairness.

Data Extraction

Bias types, evaluation methods, fairness metrics, mitigation strategies, and empirical findings.

Cross-Paper Synthesis

Identified patterns and relationships across studies to extract actionable insights.

Fairness Theory Mapping

Aligned findings with formal fairness definitions: Statistical Parity, Equal Opportunity, Disparate Impact.

How LLMs Show Bias in Hiring

1 Demographic Bias

Race, gender, disability, and nationality affect LLM ranking and selection of candidates despite merit equivalence.

2 Cultural & Linguistic Bias

LLMs penalize non-Western English, accents, and linguistic patterns, even when identity information is removed.

3 Self-Preferencing Bias

LLMs rank resumes written by AI tools higher than human-written resumes, creating tool-access inequality.

4 Amplified Human Bias

Humans follow biased AI recommendations up to 90% of the time, making human-in-the-loop an insufficient safeguard.

5 Intersectional Effects

Multiple biases compound: Black women face worst outcomes across race, gender, and occupation metrics.

Disability & Race Bias Findings

Disability Bias (Glazko et al., FAccT 2024)

GPT-4 ranks resumes with disability-related achievements lower than identical resumes without such markers.

- Autism Advocacy Award, Paralympic participation significantly reduce ranking.
- Bias severity varies by disability type.
- Custom DEI-focused GPT reduces but cannot eliminate bias.

Race & Gender Bias (Wilson & Caliskan, AIES 2024)

Resume retrieval dramatically over-selects White-coded names across 9 occupations tested.

- White names selected 85%+ of the time; Black names significantly less.
- Female names succeed only ~11% of the time in retrieval.
- Black males experience worst outcomes: intersectional bias effect.

Cultural Bias & Self-Preferencing

Cultural Bias (Rao et al., AIES 2025)

Even with anonymized identity, Indian interview transcripts scored lower than UK transcripts by GPT-4o and Gemini.

- Nationality information removed; bias persists through language alone.
- Cause: lexical density and sentence complexity differences in English usage.
- Demonstrates structural cultural embedding in LLM training data.

AI Self-Preferencing (Xu et al., AIES 2025)

LLM screening tools strongly prefer resumes written by the same model evaluating them.

- Same-model advantage ranges 68–92% in scoring.
- Candidates with GPT-4 access gain unfair advantage over manual writers.
- Simple self-recognition prompts reduce bias 60%+; introduces tool-access fairness issue.

Human-AI Interaction & Bias Amplification

Bias Amplification in Human-AI Teams (Wilson et al., AIES 2025)

When humans work alongside AI systems, they often amplify rather than correct algorithmic bias, even when aware of the AI's limitations.

Follow Biased AI Up to 90%

Humans approve biased AI recommendations at alarming rates despite expressed distrust.

Implicit Association Test Helps But Limited

Interventions reduce susceptibility but do not eliminate bias amplification.

Human-in-Loop Not Safe

Common mitigation strategy insufficient; requires systematic fairness pipeline redesign.

Benchmarking & Advanced Findings

LLMs vs Specialized Hiring Models (Anzenberg et al., 2025)

Domain-specific hiring models significantly outperform generic LLMs in both accuracy and fairness.

- Match Score achieves AUC 0.85 vs GPT-4 at 0.77 (8-point improvement).
- Specialized model shows more balanced impact ratios across demographic groups.
- Conclusion: Off-the-shelf LLMs unreliable for high-stakes hiring decisions.

Resume Bias & Generated Text (Veldanda et al., Ding et al., 2023-2024)

Resume screening reveals nuanced biases against maternity signals and political affiliation; generated text reinforces stereotypes.

- Pregnancy/maternity signals consistently reduce ranking despite legal protections.
- Political affiliation bias observed in resume analysis.
- ChatGPT-generated job text reinforces gender and socioeconomic stereotypes.

Methods for Detecting & Measuring Bias (RQ2)

Correspondence Testing

Systematically change only demographic attributes (name, gender, race, disability) while keeping experience identical to measure score differences.

Embedding & Retrieval Audits

Test LLM embeddings and retrieval bias using standardized job descriptions and controlled candidate profiles.

Human-AI Behavioral Experiments

Observe how biased AI recommendations influence human decision-making through controlled experimental trials.

Masked-LM Probing (PRISM)

Quantify bias in generated text using masked language models to measure probability distributions of biased language patterns.

Self-Preferential Audits

Compare scoring for human-written vs LLM-written resumes to identify self-preference bias in evaluation systems.

Fairness Metrics Framework (RQ3)

Your proposal metrics appear consistently across all reviewed papers. This framework ensures standardized fairness evaluation across hiring and recidivism prediction tasks.

Metric	Definition	Papers Using	Recommendation
Statistical Parity	Selection rates should be equal across demographic groups	Wilson & Caliskan, Anzenberg	Target parity ≥ 80%
Disparate Impact Ratio	Ratio of selection rates between groups	Anzenberg, EEOC Guidelines	Maintain ≥ 0.8 ratio
Equal Opportunity	Equal true positive rate for qualified candidates	Veldanda et al.	TPR balanced across groups
Error Rate Balance	False positive/negative rates consistent across groups	Multiple papers	FPR, FNR parity essential
PRISM Bias Scores	Masked-LM probability-based bias quantification	Ding et al.	Lower bias scores = fairer text

Multi-Level Mitigation Strategies

Comprehensive Mitigation Framework

No single fix eliminates LLM bias. Effective fairness requires coordinated intervention across model, pipeline, system, human, and auditing levels.

Model-Level

DEI fine-tuning, self-preference detection prompts, fairness-aware instruction tuning.

Pipeline-Level

Two-ticket robust schemes, resume summarization steps, bias filtering layers.

System-Level

Deploy domain-specific models; avoid generic LLMs for high-stakes decisions.

Human & Auditing Mitigation Strategies

Human & Auditing Safeguards

Beyond model fixes, organizations must implement human interventions and continuous fairness auditing to ensure ethical AI deployment.

Human-Level

Pre-screening interventions like Implicit Association Tests reduce susceptibility to biased AI; training essential.

Auditing-Level

Regular correspondence tests, embedding audits, MLM probing; continuous fairness monitoring post-deployment.

Governance

Establish fairness requirements, audit trails, bias incident reporting, and regulatory compliance frameworks.

LLMs are powerful but inherently risky in high-stakes domains....

THANK YOU FOR LISTENING !!

Any questions?